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Abstract

The impact of ransomware on encouraging malicious behaviour from hackers and posing a serious cybersecurity threat, examine how weaknesses in risk management led to critical disruption of essential services.

By using the WannaCry Attack Against the UK's NHS (NHS) and The Colonial Pipeline Ransomware Incident in the United States as case studies, the report will illustrate the methods by which attackers were able to successfully compromise essential systems like electronic health records, hospital systems, and fuel distribution systems by means of outdated systems, ineffective patching programs, and inadequate access controls. The discussion of this information will outline the principles of core risk management (i.e. risk identification, assessment, and control) and apply the NIST Risk Management Framework, in order to provide clear insight into how consistent application of industry best practices (e.g. timely patching, network segmentation, regular backups, and multi-factor authentication) could have lessened the overall severity of both attacks. By tying together both case examples with previously established frameworks, the report essentially establishes that to mitigate exposure to ransomware attacks it is imperative that: organizations and critical infrastructure have a comprehensive, ongoing risk management program.

1. Introduction

Risk Management/ Risk Control is an important part of business management that attempts to minimize and manage the risks an organization faces in the duration of doing business (Kenton, 2023).

Cybersecurity threats have increased significantly recently, with ransomware emerging as one of the most damaging forms of cyberattack.



Figure 1: Cybersecurity threats (Wetzel, 2024).

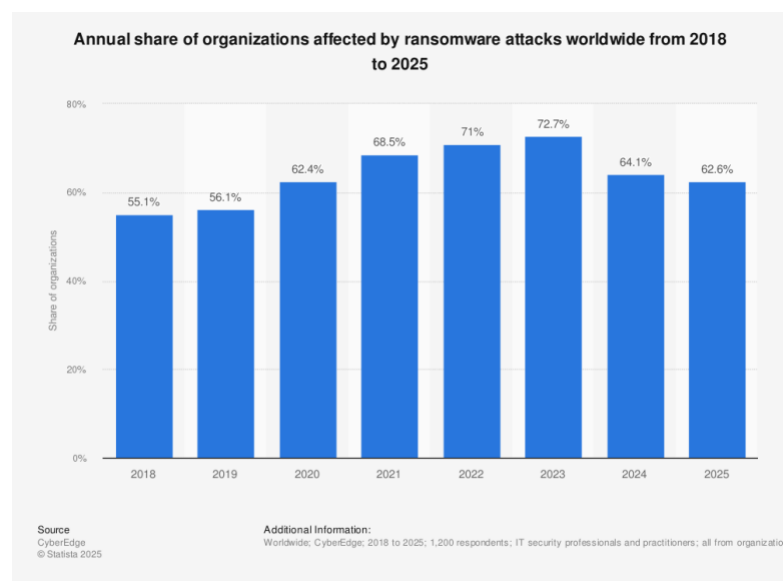


Figure 2: Statistics of Ransomware (Sjouwerman, 2023)

Ransomware is a malicious malware that encrypts critical data and systems, preventing access until a ransom is paid. Organizations with weak basic security like access control and patch management are especially vulnerable to such attacks.



Figure 3: Ransomware (Esposito, 2016).

Risk Management is important for identifying, accessing, and mitigating cybersecurity threats before they result in serious incidents. Entities, like the NIST risk management framework, provide organizations with a plan to follow for effective management of cybersecurity risks in rational and consistent manner.

This report highlights ransomware as a significant cybersecurity threat and analyses two actual ransomware events: The WannaCry Ransomware targeting the UK National Health Service and The Colonial Pipeline Ransomware in the United States. These case studies are examined from a risk management perspective to identify key vulnerabilities and evaluate how effective risk control measures could have reduced the impact of the attack. The report aims to show the importance of proactive risk management in strengthening organizational flexibility against ransomware risks.

2. Literature Review

2.1 Risk Management

The means of identifying, assessing, treating and monitoring and reporting about risk is risk management (NCSC, 2025). Risk management is not just about avoiding negative results but also about allowing positive results to help the overall success and continuity of a business (Amanda McGrath, 2025).



Figure 4: Process of risk management (BELGODERE, 2021)

2.1.1 Risk Identification

The process of identifying potential threats to an organization, its systems and its employees. It can include operations like accessing IT security threats; malware or ransomware or monitoring the weather for natural disasters and other incidents that might disrupt business operations (Amanda McGrath, 2025).

2.1.2 Risk Assessment

Risk Assessment is a systematic process of identifying hazards and assessing any related risk in a work environment, followed by the application of appropriate control measures to eliminate or mitigate them (British Safety Council, 2025).

		Risk Assessment Matrix				
Likelihood ↑	5	Medium/ High	Medium/ High	High	High	High
	4	Low / Medium	Medium/ High	Medium/ High	High	High
	3	Low / Medium	Low / Medium	Medium/ High	Medium/ High	High
	2	Low	Low	Low / Medium	Low / Medium	Medium/ High
	1	Low	Low	Low	Low / Medium	Medium/ High
		1	2	3	4	5
		Effect →				

Figure 5: Risk Assessment Matrix (Minett, 2020).

2.1.3 Risk Control



Figure 6: Risk Control (Kenton, 2023).



Figure 7: Hierarchy of Risk Control (Scott, 2024).

3. Case Study (WannaCry Ransomware)

3.1 Background

The National Health Service (NHS) was created on 5th July 1948 in the UK. Britain was the first western country to provide free healthcare to its entire population (Clement, 2023). The WannaCry ransomware attack that took place in May 2017 was a significant cyber-attack which disrupted hospital operations and affected multiple trusts nationwide.

3.2 Description of the attack

WannaCry is a kind of ransomware that exploited the EternalBlue vulnerability in Microsoft windows system (SMB – Server Message Block) Protocol (HYPR, n.d.). The cyber-attack aimed at windows PCs, encrypting vital patient data and demanded ransom in bitcoin, failure to pay would result in the attackers exposing data (NHS England, 2023). The attack spread rapidly across NHS networks, resulting in cancelled appointments and surgeries, unavailability of patient records, and temporary disruption of emergency services.

How the **WannaCry Attack** Chain Operates

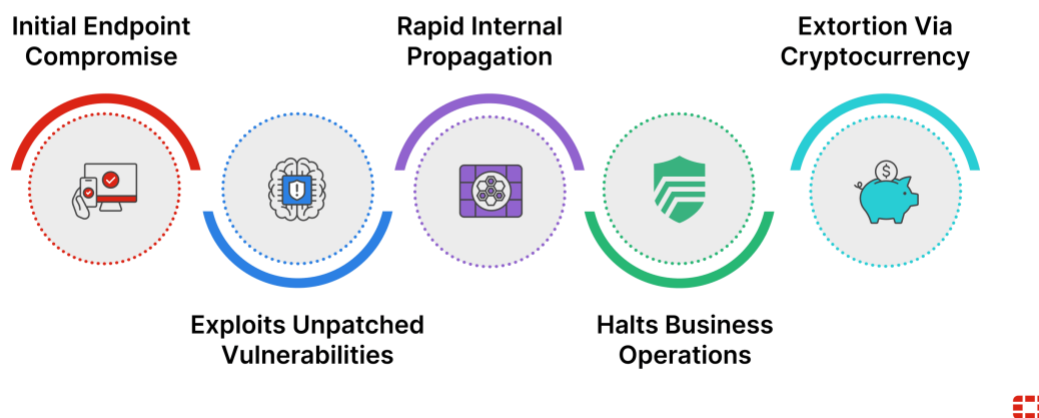


Figure 8: WannaCry ransomware chain operation (Fortinet, 2025).

3.3 Risk Identification

The process of documenting potential risks that might prevent an organization or a project from reaching its goals is risk identification (Indeed, 2022).

Assets that were at risk in the attack: Electronic Health Records, Hospital scheduling and lab systems and medical devices connected networked computers (Anon., 2024). Most infected NHS organizations were running unpatched or unsupported windows operating system, which caused the malware to infiltrate and encrypt files across networks (CLOUDFARE, 2017). The WannaCry ransomware attack was wide-ranging, affecting more than 30,000 computers across 150 countries (Ramjee, 2025).

Vulnerabilities that caused the attacks were:

1. Outdated Windows system
2. Lack of patch management
3. Insufficient network segmentation
4. Limited staff awareness



Figure 9: WannaCry Ransomware NHS (Williams, 2017).

3.3.1 Mitigation: Risk Management using NIST framework

Applying the NIST Risk Management Framework could have significantly reduced the impact of the WannaCry ransomware attack. During the categorise phase, NHS systems should have been recognised as important assets needing high security classification stage. Controls like system backups, network segmentation, and routine patch management should have been selected during the select phase. To guarantee that the vulnerabilities were fixed, these rules had to be applied to every system. Testing system defences and ensuring patch distribution would be part of the assess phase. Responsibility for risk decisions would be ensured by management approval throughout the authorisation process. Lastly, regular inspection would make sure that the systems are safe from new threats and assist in identifying new vulnerabilities (Anon., 2017).

3.3.2 Case Study Summary

The WannaCry Ransomware was discovered in 2017, infecting thousands of computers globally. A malware 'EternalBlue' created by NSA was used by the attackers identified as shadow brokers. EternalBlue and DoublePulsed (a backdoor used in the attack to install itself on the web to help propagate to other machine) enabled WannaCry to discover vulnerable systems, exploit Windows, and rapidly propagate malware across networks. The WannaCry Ransomware attack demonstrated the consequences of inadequate risk management and poor cybersecurity maintenance. Organisational risk was greatly increased by outdated systems and lack of timely patching (Liapustin, 2024).

4. Case Study 2 (Colonial Pipeline Ransomware)

The 2021 Colonial Pipeline ransomware attack highlighted inadequate cybersecurity measures, including poor network segregation, weak authentication, and slow vulnerability fixes, emphasizing the need for operational resilience and better coordination with cybersecurity agencies. A more detailed analysis of this incident is provided in [Appendix Case Study \(Colonial Pipeline Ransomware\)](#).

4. Conclusion

Both WannaCry and Colonial Pipeline case studies illustrate that while attacker sophistication contributed to their respective attacks, far less damage could have been inflicted upon both systems with good working knowledge of how to manage basic security controls, such as patching systems, implementing strong access controls, controlling patching, etc. Both operations would have been able to limit the negative effects of these two incidents if they had:

Considered their core operational systems to be high-risk assets and used structured frameworks, like NIST, which include controls that would require an organisation to perform continual processes in accordance with best practices to prevent disruptions from such attack methods as ransomware; and

Implemented a well-defined governance model for managing risk, including the use of process-based technical controls, adequate training of personnel and a defined level of accountability for each person in the organisation.

Thus, to effectively sustain a high level of critical service delivery, and mitigate operational, financial and reputation risks stemming from ransomware attacks on their organisation, an organisation must clearly establish a solid foundation of continuous operational excellence through proper processes for the management of both technical and staff.

5. Bibliography

Clement, M., 2023. *history.blog.gov.uk*. [Online]

Available at: <https://history.blog.gov.uk/2023/07/13/the-founding-of-the-nhs-75-years-on/>
[Accessed 20 12 2025].

NHS England, 2023. *NHS England business continuity management toolkit case study: WannaCry attack*. [Online]

Available at: <https://www.england.nhs.uk/long-read/case-study-wannacry-attack/>
[Accessed 20 12 2025].

HYPR, n.d. *What is EternalBlue? | Security Encyclopedia*. [Online]

Available at: <https://www.hypr.com/security-encyclopedia/eternalblue>
[Accessed 20 12 2025].

Fortinet, 2025. *What is WannaCry? WannaCry Ransomware Attack Case Study*. [Online]

Available at: <https://www.fortinet.com/resources/cyberglossary/wannacry-ransomware-attack>
[Accessed 20 12 2025].

Indeed, 2022. *What Is Risk Identification? Definition and Tools*. [Online]

Available at: <https://www.indeed.com/career-advice/career-development/risk-identification>
[Accessed 20 12 2025].

Anon., 2024. *Investigation: WannaCry cyber attack and the NHS - NAO report*. [Online]

Available at: https://www.nao.org.uk/reports/investigation-wannacry-cyber-attack-and-the-nhs/?utm_source=chatgpt.com
[Accessed 20 12 2025].

CLOUDFARE, 2017. *What was the WannaCry ransomware attack?*. [Online]

Available at: <https://www.cloudflare.com/en-gb/learning/security/ransomware/wannacry-ransomware/>
[Accessed 10 12 2025].

CLOUDFARE, 2017. *What was the WannaCry ransomware attack?*. [Online]

Available at: <https://www.cloudflare.com/en-gb/learning/security/ransomware/wannacry-ransomware/>
[Accessed 20 12 2025].

Ramjee, D., 2025. *From ransoms to ruin: Are extortion payments by ransomware victims insurable?*. [Online]

Available at:
https://www.researchgate.net/publication/398003232_From_ransoms_to_ruin_Are_extortion_payments_by_ransomware_victims_insurable
[Accessed 20 12 2025].

Anon., 2017. *Search Results | NIST*. [Online]

Available at: <https://www.nist.gov/search?s=wannaCry+ransomware>
[Accessed 23 12 2025].

Liapustin, M., 2024. *Trustifi*. [Online]

Available at: <https://trustifi.com/blog/wannacry-ransomware-attacks-propagating-through-email-case-study/>
[Accessed 23 12 2025].

Easterly, J., 2023. *The attack on colonial pipeline: What we've learned & what we've done over the past two years*. [Online]

Available at: <https://www.cisa.gov/news-events/news/attack-colonial-pipeline-what-weve-learned-what-weve-done-over-past-two-years>
[Accessed 2025].

Fanning, J. E. T., 2023. *The attack on colonial pipeline: What we've learned & what we've done over the past two years*. [Online]

Available at: <https://www.cisa.gov/news-events/news/attack-colonial-pipeline-what-weve-learned-what-weve-done-over-past-two-years>
[Accessed 23 12 2025].

WALLIX, 2023. *What Happened in the Colonial Pipeline Ransomware Attack - WALLIX*. [Online]

Available at: <https://www.wallix.com/blogpost/what-happened-in-the-colonial-pipeline-ransomware-attack-2/>
[Accessed 23 12 2025].

Huntress, 2015. *Colonial Pipeline Ransomware Attack: Impact, Victims, Recovery | Huntress*. [Online]

Available at: <https://www.huntress.com/threat-library/ransomware/colonial-pipeline-ransomware>
[Accessed 24 12 2025].

Altman, W., 2021. *Cyber catastrophe meets enterprise ransomware: Colonial Pipeline attack*. [Online]

Available at: <https://insights.cybcube.com/en/colonial-pipeline-attack>
[Accessed 24 12 2025].

FortiGuard Labs, 2021. *Protecting Critical Infrastructure: Colonial Pipeline, DarkSide, and Ransomware | FortiGuard Labs*. [Online]

Available at: <https://www.fortinet.com/blog/threat-research/protecting-critical-infrastructure-colonial-pipeline-darkside-and-ransomware>
[Accessed 24 12 2025].

Wetzel, E., 2024. *What are the 8 main cyber security threats? - FortifyData unified cyber risk & asset intelligence platform*. [Online]

Available at: <https://fortifydata.com/blog/what-are-the-8-main-cyber-security-threats/>
[Accessed 24 12 2025].

Kaspersky, 2019. *What Is Cyber Security?*. [Online]
Available at: <https://www.kaspersky.com/resource-center/definitions/what-is-cyber-security>
[Accessed 24 12 2025].

Esposito, J., 2016. *Expert tips to avoid falling victim to ransomware*. [Online]
Available at: <https://www.kaspersky.com/blog/expert-ransomware-tips/11974/>
[Accessed 24 12 2025].

NCSC, 2025. *Search Results*. [Online]
Available at: <https://www.ncsc.gov.uk/search?q=risk%20management>
[Accessed 25 12 2025].

Amanda McGrath, A. J., 2025. *Risk Mangement*. [Online]
Available at: <https://www.ibm.com/think/topics/risk-management>
[Accessed 25 12 2025].

Amanda McGrath, A. J., 2025. *Risk Management*. [Online]
Available at: <https://www.ibm.com/think/topics/risk-management>
[Accessed 25 12 2025].

Kenton, W., 2023. *Risk Control*. [Online]
Available at: <https://www.investopedia.com/terms/r/risk-control.asp>
[Accessed 26 12 2025].

Sjouwerman, S., 2023. *73% of Organizations Affected by Ransomware Attacks Globally in 2023, According to Statista*. [Online]
Available at: <https://blog.knowbe4.com/ransomware-attacks-affect-majority-organizations-2023>
[Accessed 26 12 2025].

Minett, A., 2020. *The 5 Steps To Risk Assessment Explained*. [Online]
Available at: <https://www.chas.co.uk/blog/5-steps-to-risk-assessment/>
[Accessed 26 12 2025].

British Safety Council, 2025. *Risk Assessment and Management: A Complete Guide | British Safety Council*. [Online]
Available at: <https://www.britsafe.org/training-and-learning/informational-resources/risk-assessments-what-they-are-why-theyre-important-and-how-to-complete-them>
[Accessed 26 12 2025].

Scott, D., 2024. *How to Apply the Hierarchy of Controls to Different Types of Risks*. [Online]
Available at: <https://safetymanagementgroup.com/blog/applying-the-hierarchy-of->

controls-to-different-types-of-risks/

[Accessed 26 12 2025].

Kenton, W., 2023. *Risk Control*. [Online]

Available at: <https://www.investopedia.com/terms/r/risk-control.asp>

[Accessed 26 12 2025].

BELGODERE, M., 2021. *The Risk Management Process: 4 Essential Steps*. [Online]

Available at: <https://www.migso-pcubed.com/blog/risk-management/four-step-risk-management-process/>

[Accessed 29 12 2025].

Williams, D., 2017. *WannaCry Ransomware and the NHS attack*. [Online]

Available at: <https://www.blackfog.com/wannacry-ransomware-nhs-attack/>

[Accessed 20 01 2026].

Garber, J., 2021. *Colonial Pipeline fiasco foreshadows impact of Biden energy policy*. [Online]

Available at: <https://www.foxbusiness.com/markets/colonial-pipeline-fiasco-foreshadows-impact-of-biden-energy-policy>

[Accessed 20 12 2026].

dslk-dev, 2022. *The Colonial Pipeline Ransomware Attack, One Year Later | Analyst1*. [Online]

Available at: <https://analyst1.com/the-colonial-pipeline-ransomware-attack-one-year-later/>

[Accessed 20 12 2026].

7. Appendix Case Study (Colonial Pipeline Ransomware)

7.1 Background

In May 2021, the Colonial Pipeline ransomware attack struck one of the biggest fuel pipeline operators in the U.S. (Fanning, 2023). Colonial Pipeline provides operators in the U.S.. After the assault, the firm ceased its pipeline activities to limit the damage, resulting in fuel scarcity and widespread panic. The attack was attributed to the Darkside (eastern European cybercriminal organization) ransomware group and highlighted the cybersecurity risks faced by critical infrastructure organisations, particularly those with weak access control and inadequate risk management practices (WALLIX, 2023).

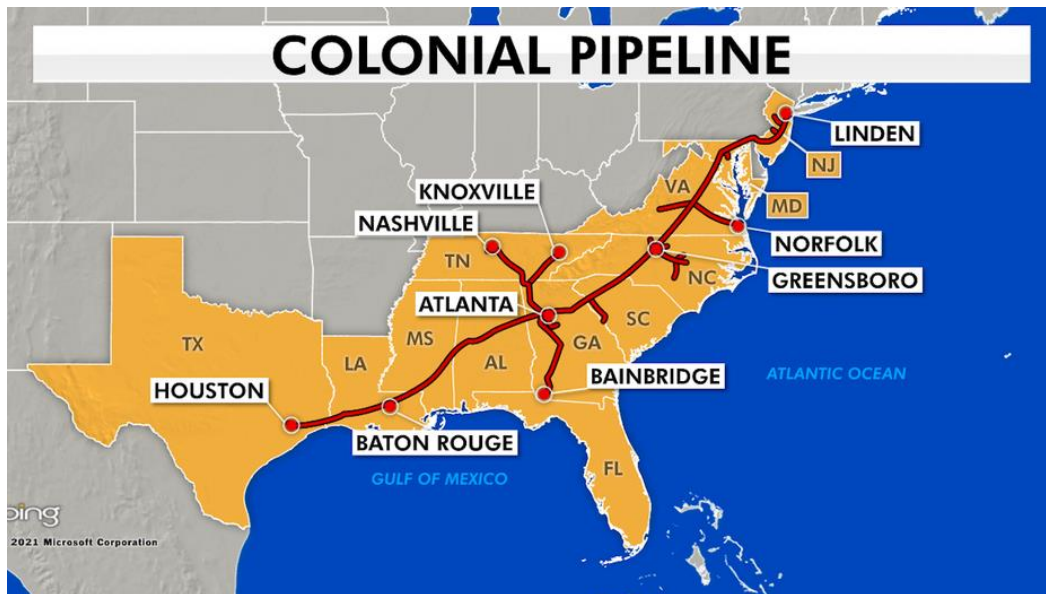


Figure 10: Colonial Pipeline distributing areas (Garber, 2021).

7.2 Description of the attack

The attackers accessed colonial pipeline's corporate network through compromised credentials, allegedly because there was no multi-factor authentication (MFA) enabled on a remote access account. Once inside the network, the DarkSide ransomware was deployed, encrypting important data systems. While the operational technology systems remained unaffected, the company decided to halt pipeline operations as a safety measure. To regain access quickly, Colonial Pipeline paid a ransom of approximately 64 Bitcoin as ransom. Due to fluctuating value of cryptocurrency, the recovered bitcoins were worth around \$2.4million.

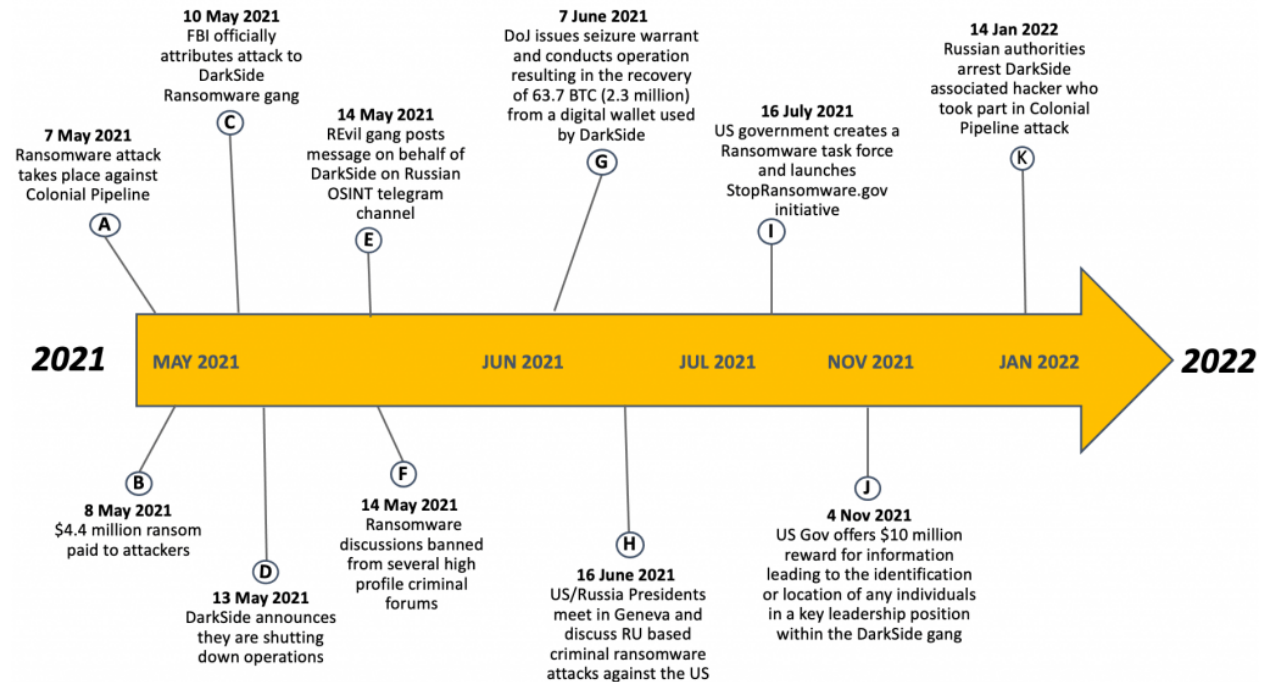


Figure 11: Timeline of Colonial Pipeline Ransomware attack (dslk-dev, 2022).

7.2.1 Risk Identification

The Colonial Pipeline event uncovered various serious cybersecurity vulnerabilities.

Assets at risk were IT systems, billing systems and operational continuity of fuel distribution. Threats of the attack was ransomware encrypting essential business systems.

Vulnerabilities that caused the attacks were:

1. Absence of multi-factor authentication
2. Ineffective credential management
3. Insufficient network management

The impacts caused to the Colonial Pipeline were Pipeline shutdown, fuel shortages, financial loss, reputational damage, and national security concerns (Altman, 2021).

7.2.2 Mitigation and Prevention Strategies

To mitigate the possible incoming risks attacks like Colonial Pipeline and organizations should:

1. Use MFA: Require multi-factor authentication for all important accounts or devices to prevent unauthorized access.
2. Patch and Update: Keep track of known vulnerabilities on a frequently to prevent attacks.
3. Train Employees: Educate staff on identifying Phishing attempts and other threat areas.
4. Backup Data: Maintain offline and manual versions of data to ensure fast recovery in case of attacks (Huntress, 2015).

7.5 Case Study Summary

On Thursday, May 6th, the leading U.S. fuel pipeline operator, Colonial Pipeline was targeted by a ransomware attack by Darkside according to FBI. They gained access to the network using compromised data attempted to steal nearly 100GB of critical data, locked down the data and servers and asked to pay ransom. These events caused Colonial Pipeline to shut all their Pipelines, fuel shortages, financial loss, reputational damage, and national security concerns. This incident highlights the importance of strong risk management practices and reinforces the need for continuous monitoring (FortiGuard Labs, 2021).